



Layer 2 switch attacks

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Purpose:

The purpose of this lab was to familiarize ourselves with layer 2 as well as some basic attacks that are used to gain access to unprotected switches. We then learned how to mitigate these attacks or why we do some mitigation commands when we configure switches. While the main purpose of the lab was layer 2 attacks, we also got lots of experience with using virtual machines, Linux, and topology troubleshooting, which is arguably just as important if not more important of a skill to learn.

Background:

Layer 2 or the Data Link Layer is the second layer in the OSI model, which has seven layers. Layer 2's job is to move data across a local network between two devices. Layer 2 uses Mac addresses to identify devices and send data on the local network. Layer 2 groups data into something called frames and forwards the frames across the network; these frames provide more structure as layer 2 is not just forwarding random clumps of data across the network. Layer 2 also has some redundancy in the form of a cyclic redundancy check (CRC). CRC will drop frames that get corrupted. Layer 2 importantly does not resend or correct the corrupted frames as this correction occurs in higher layers.

Why Linux?

All of the layer 2 attacks that we performed were done on Linux. Linux is an open-source operating system that is separate from Windows. Because Linux is open source, you can control, customize, and change whatever you want about the OS. Linux also has different distributions which are people who have taken the Linux code and slightly tweaked it and/or added packages or drivers. They are premade versions of Linux. These days there are hundreds of distros to choose from which are all slightly different flavors of Linux. While they all have slightly tweaked Linux, they all still use the same kernel, which is the Linux kernel as well as GNU. Kali Linux, a Linux distribution which is based on Debian, has many different premade attacks

tools which have been added and will come preinstalled with the distro which can be used for these attacks. Kali is a distro of Linux that is specifically made attacking and is a basic beginner distro made for people learning about cybersecurity. The preinstalled tools are fairly straight forward and provides a UI for most of their tools. For our Arp spoofing attack, we chose to use Kali Linux. It has a tool called dsniff which has tools to do an Arp spoofing attack. I also wanted to diversify the distros that we used. I choose to use Ubuntu for the Macof attack because it is a well know and trusted distro of Linux that is very stable using Debian like Kali. Using Ubuntu for attacking poses a little more difficulty than using a UI like what is on Kali. On Debian based distros you can use apt to install apps though because you must install the attacking tools yourself as they do not come preinstalled with the distro. While Ubuntu does come with some preinstalled tools as well as drivers, Ubuntu is more of a general-purpose distro that is fairly simple to use.

DHCP

One of our attacks was a DHCP starvation attack. This usually happens when so many fake requests are sent that the server runs out of available IP addresses in its pool. DHCP (Dynamic Host Configuration Protocol) gives a device that connects to a network with an IP address so that they can connect to the internet. DHCP allows you to be convenient. DHCP works on a range of sizes of networks. Almost all routers except to not have to statically set an IP address on a device, which makes it much more ones that are really old have DHCP capability. DHCP also supports both 1pv4 and ipv6.

DHCP was originally developed in 1993 and was revolutionary as people no longer needed to statically set up their own IP address. DHCPv6 was then added in 2003 which gave support for ipv6 with the diminishing IP addresses in the early 21st century, although DHCP can generally be used for addressing private ip addresses client sends a DHCP request which is a broadcast. After the request is sent, any DHCP server which is on the network may process the request. A DHCP server is configured to send out a lease for the device. The Lease is renewed periodically, which can be by an admin and will send out the specific configurations. The DHCP server will then be configured in the DHCP server. The actual operation between the client server goes: DISCOVER, OFFER, REQUEST, ACKNOWLEDGE.

Arp spoofing

Another attack that we did was an Arp spoofing which is a man in the middle attack. Usually, the victims will just send data to the router, but a man in the middle will as the name suggests place themselves in between the victim and the router and intercept the data. From there they are free to sniff the data and steal whatever they want. They do this by associating themselves with another host and using the same MAC address as the host. This is one of the reasons that good encryption is so important because it is a layer of protection between yourself and attackers. While attackers can sniff data on the victim since they are in the middle, they can do a lot more than this. Attackers can also perform a denial-of-service attack. This is where the attacker prevents any data from passing the router so that it can go out to the internet. Attackers can demand a payment which can sometimes be millions of dollars for large companies. The Attacker can also modify traffic that they want to if the data is unencrypted and steal any passwords or banking information. It is very important to use a VPN while on public Wi-Fi where you will most often see people doing something like this. A VPN will connect to a different server and create a private tunnel that is encrypted. While this does protect you from attackers on Wi-Fi, it does not protect you from things on the internet. Another thing that you should be using is HTTPS and SSH rather than HTTP and telnet. These protocols provide encryption which will provide a layer of protection although you are not immune to attacks. DAI (Dynamic ARP Inspection) will confirm that a host is who they say they are and will shut down someone who is a man in the middle as they would have a different Ip address.

Lab summary

In this lab we had to do three attacks on switches and the attacks were all on layer 2. I choose to do ARP spoofing, macoff, and DHCP snooping. These were all performed on Linux, and they were done on the same topology of one router, one switch, and two pcs. We used tools from Kali Linux and from dsniff in order to perform the attacks. For the first two attacks we used virtual machines as they act like a third host which we needed and allowed us to use windows which is what we currently had installed. We used an actual Linux distro for the third as we dual booted Linux mint and Windows 11. This was just easier for us to use so that we didn't have to mess with virtual machines for another attack.

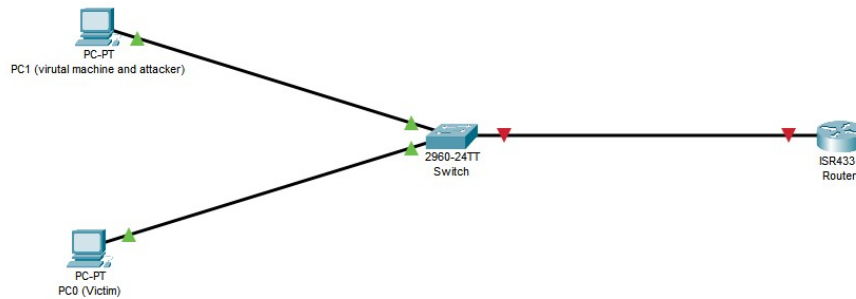
Lab Commands

Switchport port-security: applies port security to the configuration uses shutdown violation to start with.

Switchport port-security max: limits the number of MAC addresses before the violation occurs.

Switchport port-security violation: allows you to configure violations.
Protect, restrict, shutdown (recommended)
Ip dhcp snooping: used to protect against rogue DHCP servers and
prevents some attacks. Filters traffic.

Network topology



Configs

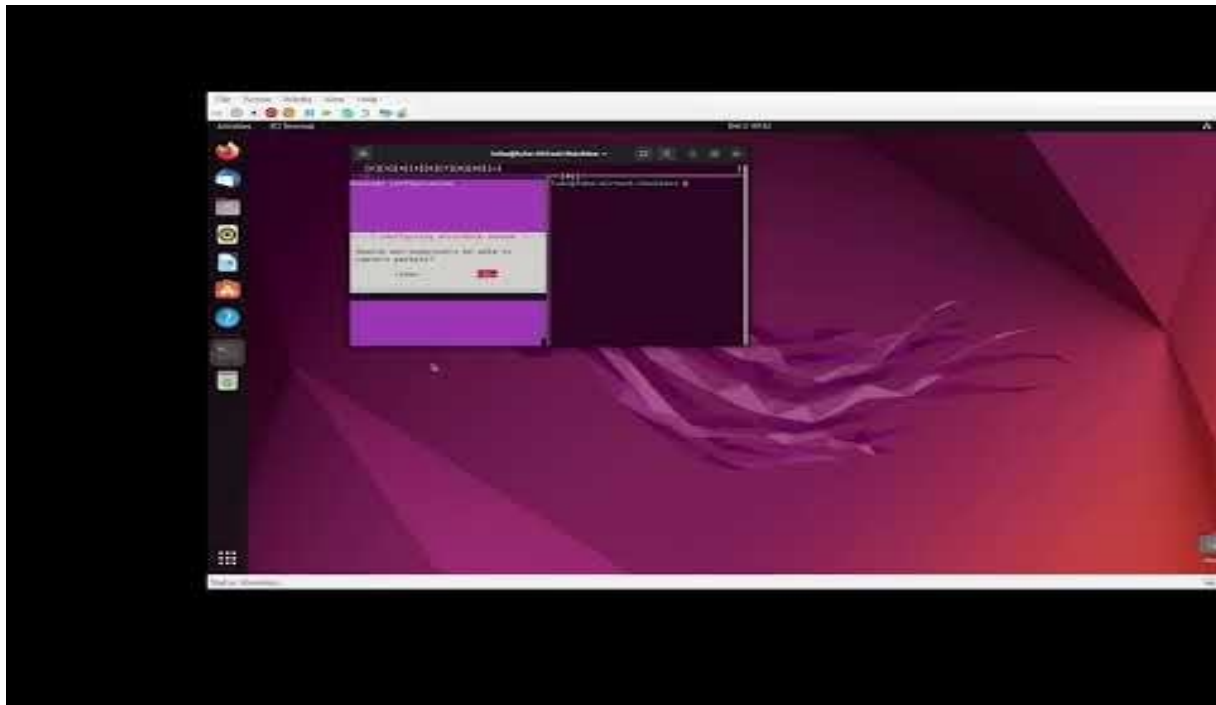
```
sudo systemctl restart isc-dhcp-server  
sudo systemctl enable isc-dhcp-server
```

```
# Check for config errors  
sudo dhcpd -t -cf /etc/dhcp/dhcpd.conf
```

```
Router(config)# ip dhcp pool DOG.com  
Router(dhcp-config)# network 192.168.1.0 255.255.255.0  
Router(dhcp-config)# default-router 192.168.1.1  
Router(dhcp-config)# dns-server 8.8.8.8  
Router(dhcp-config)# domain-name dog.lan
```

Macof:

<https://www.youtube.com/watch?v=ODH2j2C2EIs&feature=youtu.be>



ARP Spoofing:

Either set up a virtual machine and virtually run kali Linux in a virtual environment or what I suggest partitioning your drive and temporarily installing kali. These instructions were given from that assumption.

```
Ethernet adapter Ethernet:
Connection-specific DNS Suffix . : 
Description . . . . . : Realtek PCIe 2.5GbE Family Controller
Physical Address. . . . . : 58-11-22-4B-E0-8D
DHCP Enabled. . . . . : yes
Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . . : fe80::e3b9:4d5c:12a6:c200%12(Preferred)
IPv4 Address. . . . . : 192.168.221.116(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Lease Obtained. . . . . : Saturday, March 7, 2026 12:48:31 AM
Lease Expires . . . . . : Saturday, March 14, 2026 12:48:28 AM
Default Gateway . . . . . : 192.168.221.99
DHCP Server . . . . . : 192.168.221.99
DHCPv6 IAID . . . . . : 106434850
DHCPv6 Client DUID. . . . . : 00-01-00-01-2A-C1-67-92-58-11-22-4B-E0-8D
DNS Servers . . . . . : 75.75.75.75
                          75.75.76.76
NetBIOS over Tcpi. . . . . : Enabled
```

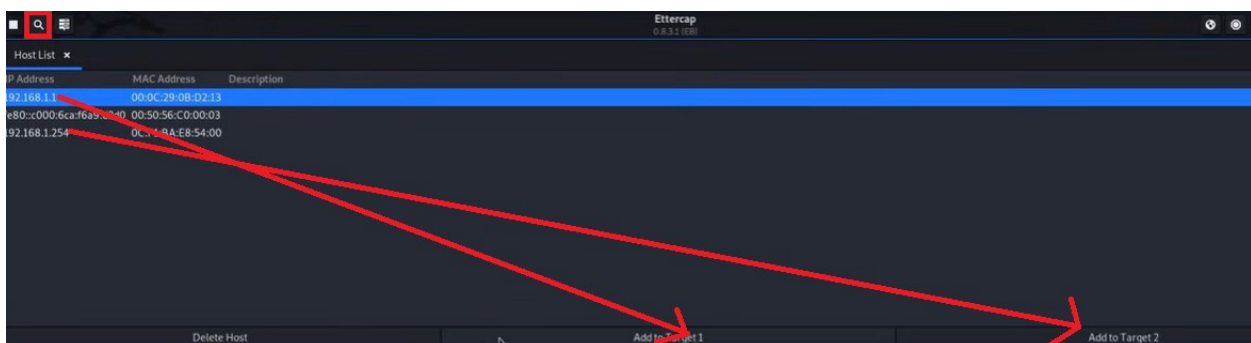
Do an ipconfig/ all or and ip a if on Linux and locate your ip address and MAC address (physical address). Make note of these for later.

Once in kali navigate to the tools tab and search for ettercap-graphical and click on it. Enter your password.

If you are not using Kali and another distro use the command `sudo apt-get install -y ettercap-graphical` to install ettercap



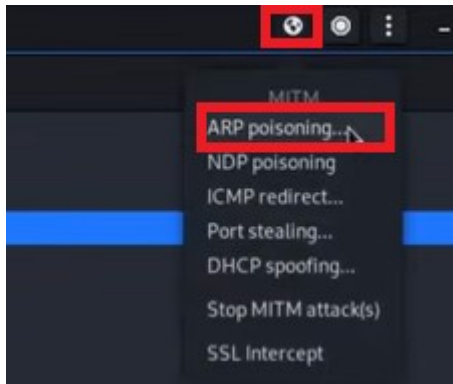
Once into Ettercap choose the interface that you are using like eth0 and click the check at the top



Click the search icon in the top left and it will scan the subnet for devices, and it will give you a list of devices

After the scan is complete add the victim's IP to target one

Open Wireshark and choose the interface that you used earlier



Go back to ettercap and find the attacking tab at the top and click on the arp poisoning and click ok

Go back to wireshark and you should be able to go on the victim's and search for something and see it on the attackers' pc.

To stop the attack, you can click on the stop button at the top next to the attack tab.

To mitigate the ARP Spoofing attack, it is fairly simple. You first need to get into your switch. Then enter these basic commands in.

Enable DHCP Snooping:

```
Switch(config)# ip dhcp snooping  
Switch(config)# ip dhcp snooping vlan 1
```

Then set the port connected to DHCP server/router to trust:

```
Switch(config)# interface fa0/24  
Switch(config-if)# ip dhcp snooping trust
```

Enable DAI:

```
Switch(config)# ip arp inspection vlan 1
```

Then set routers and other switches that are currently on the network to trust: Switch(config)# interface fa0/24

```
Switch(config-if)# ip arp inspection trust
```

1. Type the following commands to set up the network for basic DHCP configuration as it is required for DHCP starvation.

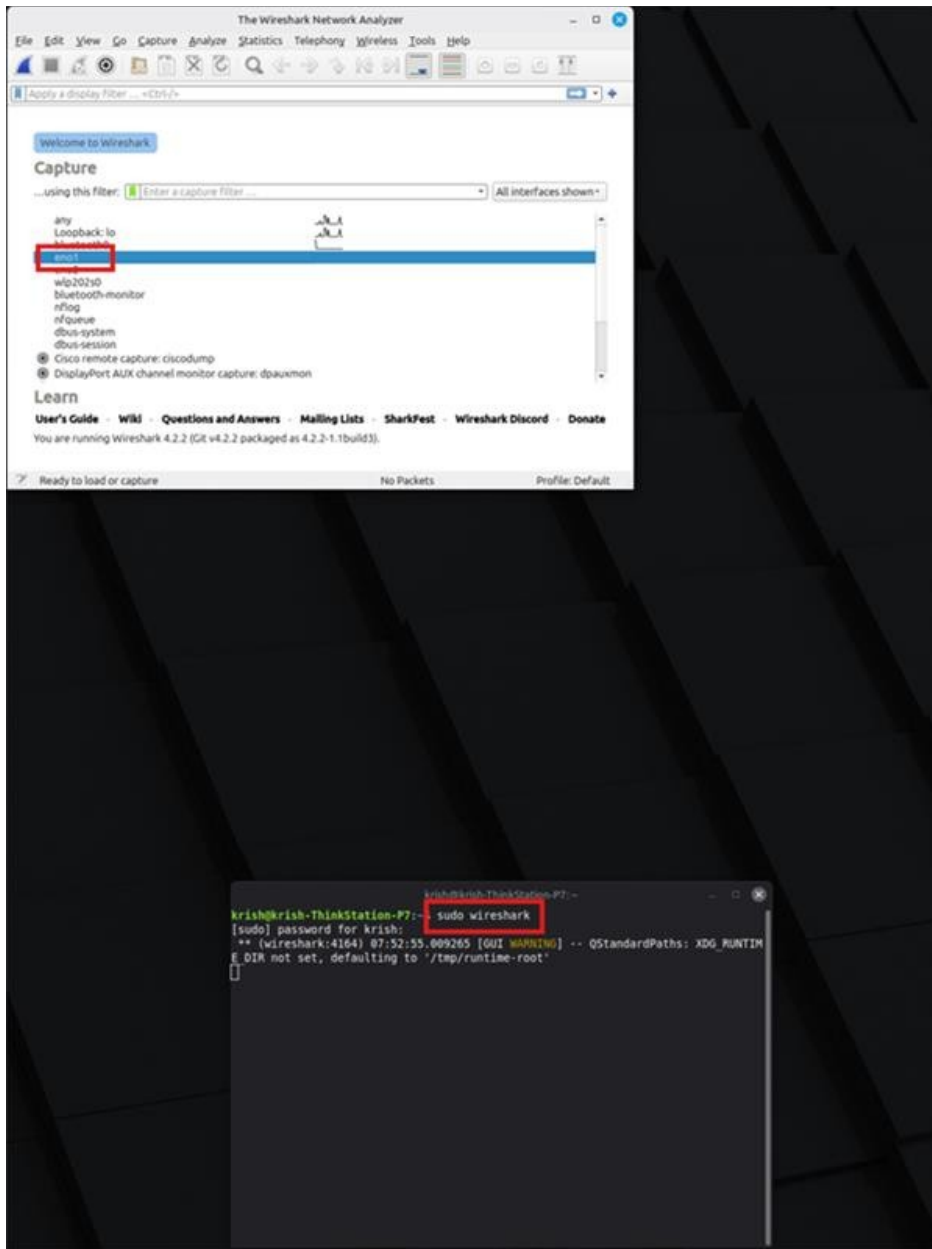
```
ip dhcp pool LAN
```

```
network 192.168.1.0 255.255.255.0
```

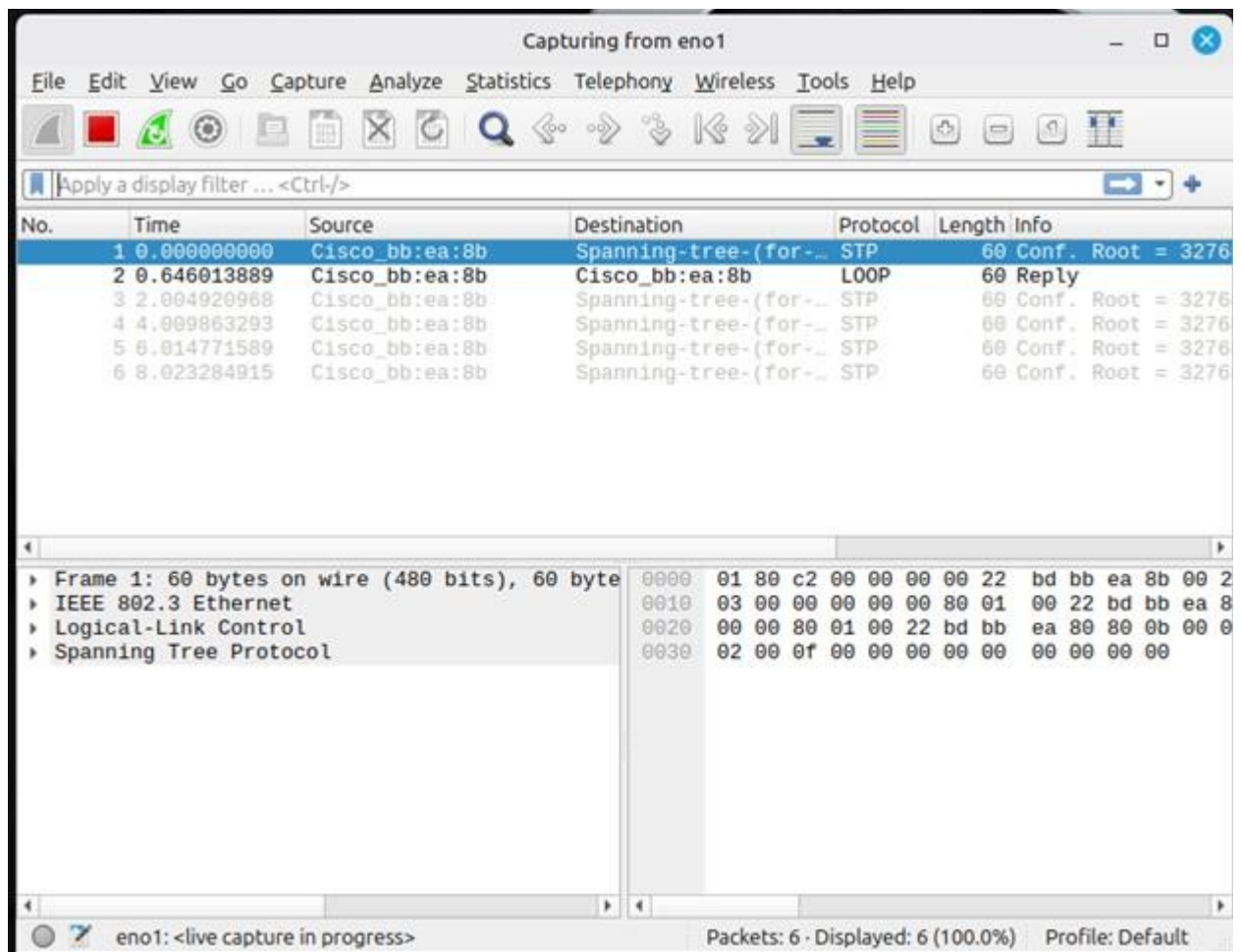

default-router 192.168.1.1

dns-server 8.8.8.8

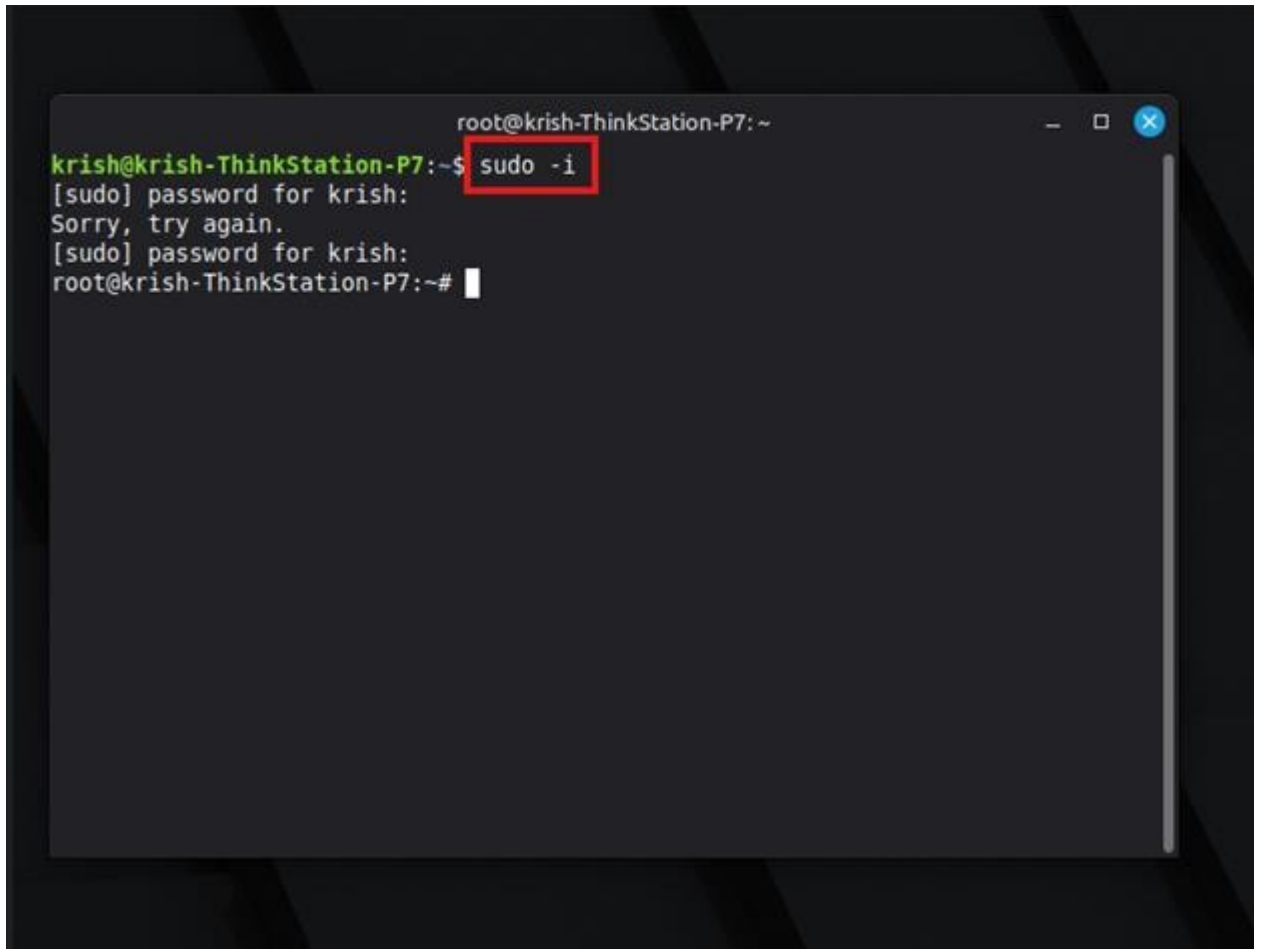
domain-name local.lan



2. Use the command `sudo apt install -y wireshark` if wireshark is not install then use the `sudo wireshark` command in terminal or press your super key and search for “wireshark” and select the interface that you are using as highlighted in the image above once it opens.

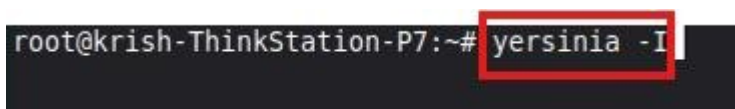


3. The wireshark will open and you should see something similar to the image above, or if you did not previously have wireshark installed onto your computer, follow the guide in the Windows 11 setup guide.



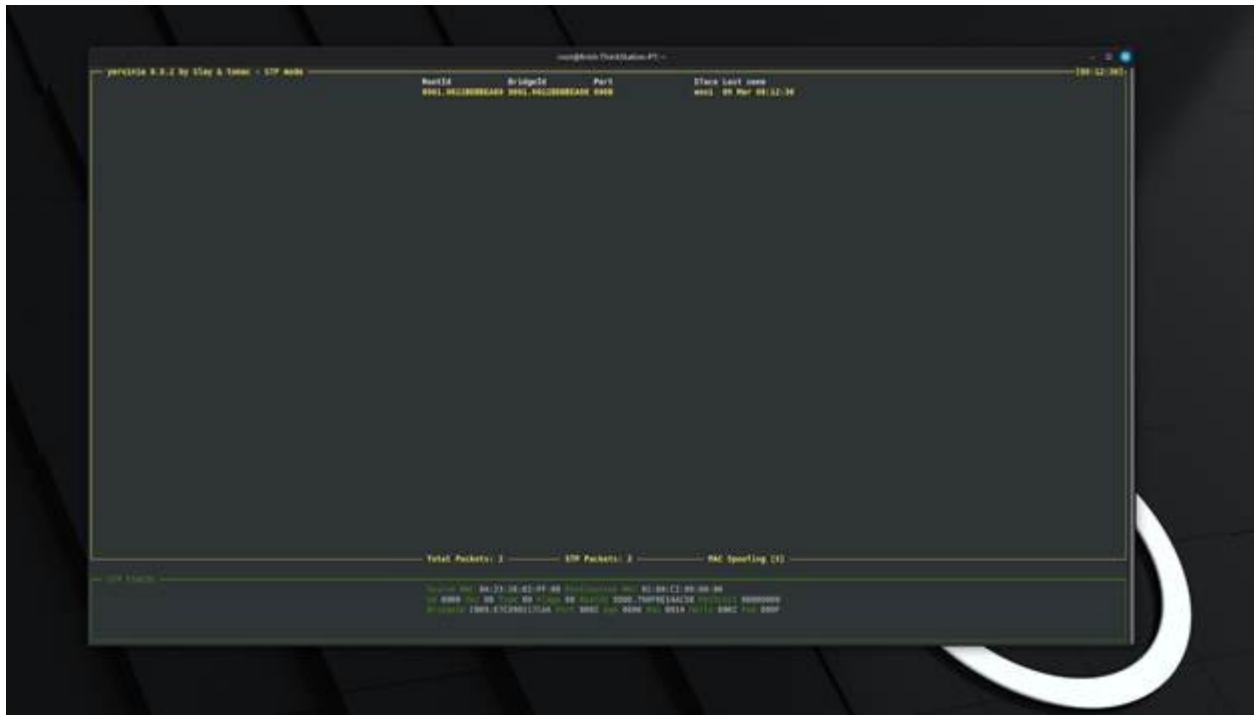
```
root@krish-ThinkStation-P7: ~  
krish@krish-ThinkStation-P7:~$ sudo -i  
[sudo] password for krish:  
Sorry, try again.  
[sudo] password for krish:  
root@krish-ThinkStation-P7:~#
```

4. Type the *sudo -i* command as highlighted in the image above.

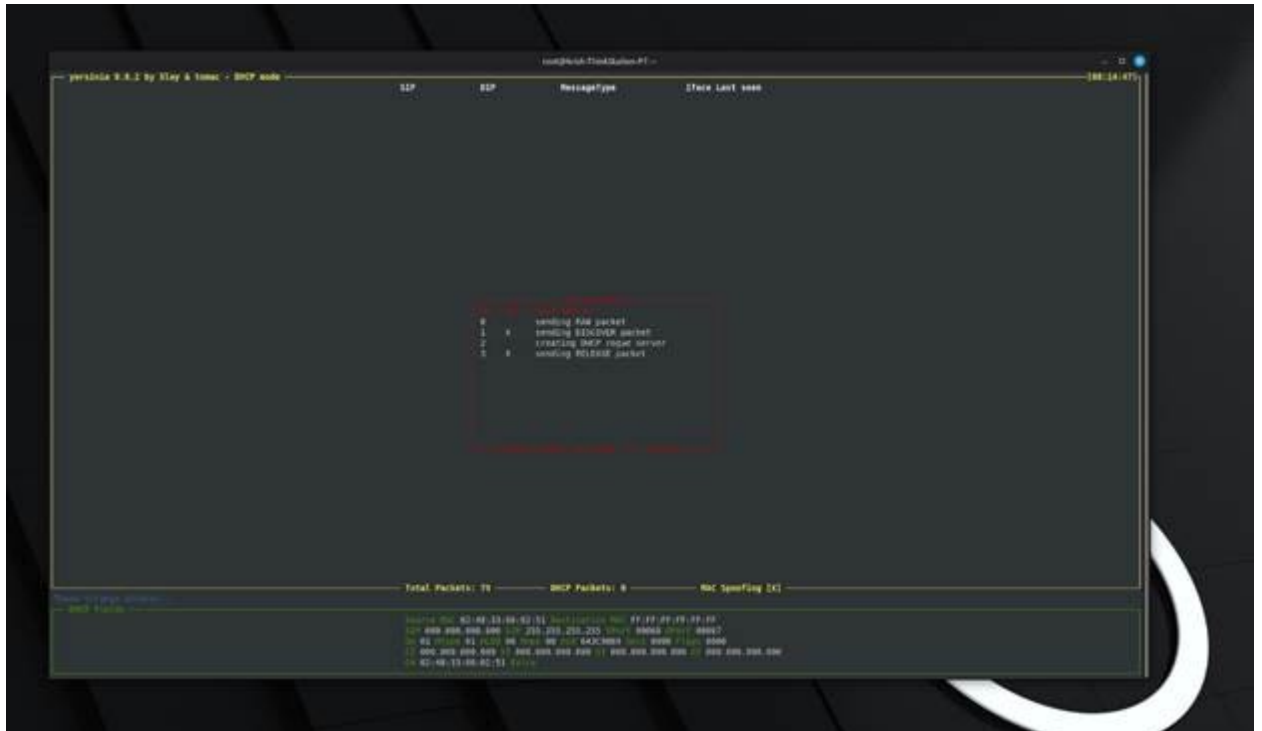


```
root@krish-ThinkStation-P7:~# yersinia -I
```

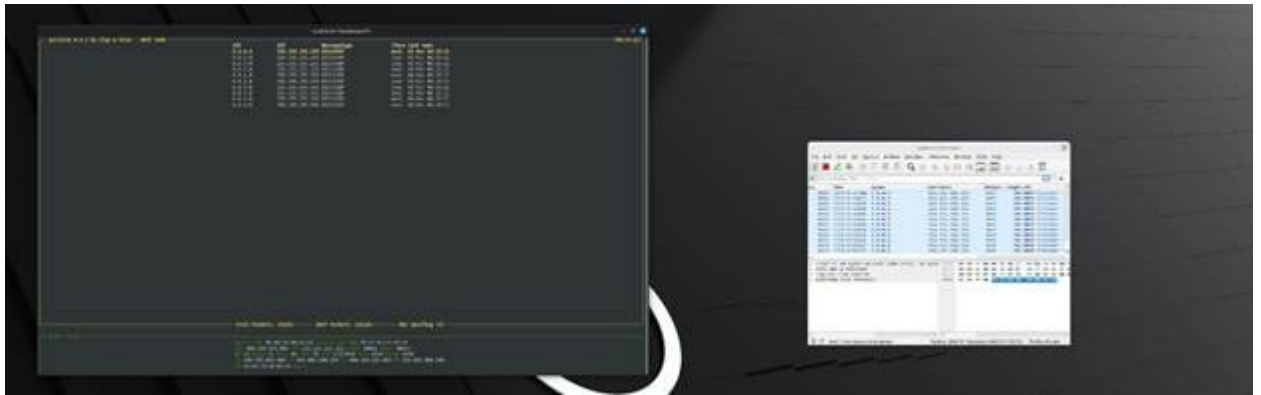
5. Type the *yersinia -I* command as highlighted in the image.



6. Once the yersinia application opens as shown in the image above, press F2 on your computer to advance to the next step.



7. Next, press the letter “x” on your computer and then the number “1” and then the letter “q” in order to send out DHCP Discover packets to the victim PC.



8. After you complete step 7, the Wireshark application should be constantly showing DHCP Discover packets and the yersinia application should be showing the attack working as well as the time the attack started. Next, I will be demonstrating you how to mitigate a DHCP starvation attack.

```
COM1 - PuTTY
eliIsbettertho#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
eliIsbettertho(config)#ip dhcp snooping
eliIsbettertho(config)#ip dhcp snooping vlan 1
eliIsbettertho(config)#int f0/10
eliIsbettertho(config-if)#ip dhcp snooping trust
eliIsbettertho(config-if)#exit
```

```
eliIsbettertho(config)#int ra fa0/1-9 , fa0/11 - 24
eliIsbettertho(config-if-range)#ip dhcp snooping limit rate 5
eliIsbettertho(config-if-range)#switchport mode acc
eliIsbettertho(config-if-range)#switchport port-security
eliIsbettertho(config-if-range)#
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#do clear ip dhcp binding
% Incomplete command.

Router(config)#do clear ip dhcp binding ?
*          Clear all automatic bindings
A.B.C.D    Clear a specific binding
vrf        DHCP vrf bindings

Router(config)#do clear ip dhcp binding *
Router(config)#
```

9. On the Switch (name: elilsbettertho) and Router type all the following commands in both images above to prevent a DHCP Starvation attack from occurring on your network.

Problems:

This lab took a lot more of our time than we would have like because we spent a lot of time in our lab just troubleshooting mainly the topology and correctly setting up the virtual machine. Many of the issues were that the ethernet that is tucked way back behind our desk was not working somewhere and we didn't know what was wrong. We did trace it to an ethernet extender that was unplugged, so we plugged it back in and it worked for now. We continued the lab and set up a virtual machine, but in order for a virtual machine to be able to ping out you have to bridge it you should have ethernet as it is a much easier solution, and our ethernet was not working again and we were just using wireless. After troubleshooting we found that the ethernet cables in the back of the computer were changed around to the wrong ports. While we did fix the ethernet, we choose to just install our linux distro on a drive as it would be easier in the future to use and if our ethernet breaks again then we would not have to fix it and we could just use wireless. It seemed like someone who was not us was switching around the ethernet cables for their use and in the process unplugged other cables that broke our rack. When doing the macof attack we ran into some issues with using the terminal as I was struggling to use the correct commands for the attack. Many of the people explaining the attack online were using kali Linux where everything just uses a UI. Overall this lab took a while for several reasons other than actually doing the attacks themselves, and even though we spent a lot of time troubleshooting I still think that we learned important things like how to setup a virtual machine and how to setup Linux.

Conclusion:

This lab taught us a lot about how linux works as well as virtual machines. We learned basic linux terminal commands in Debian and how to use them, and we learned the benefits of linux and when we should use linux which we can use in future lab. We also learned how to do the actual layer 2 attacks on both a terminal and using a UI.